

Work Breakdown Structure(WBS)

Project: RollCall - Face recognition attendance system

1. Project Planning & Requirements

- 1.1 Define project goals and objectives
- 1.2 Identify stakeholders and user roles
- 1.3 Gather functional requirements
- 1.4 Gather non-functional requirements
- 1.5 Define system scope and constraints
- 1.6 Create Software Requirements Specification (SRS)
- 1.7 Review and finalize requirements

2. System Design

- 2.1 Define overall system architecture
- 2.2 Design database structure
- 2.3 Design user interface layouts
- 2.4 Define attendance workflow
- 2.5 Define facial recognition workflow
- 2.6 Design privacy and security model
- 2.7 Create system diagrams

3. Face Recognition Module

- 3.1 Research and select face recognition library
- 3.2 Define student enrollment process
- 3.3 Design face detection process
- 3.4 Define face matching logic
- 3.5 Define recognition confidence threshold
- 3.6 Define error handling for unrecognized faces

4. Attendance Management System

- 4.1 Define attendance session process
- 4.2 Define attendance statuses (Present, Late, Absent)
- 4.3 Design instructor dashboard features
- 4.4 Define manual attendance override process
- 4.5 Define attendance record storage rules
- 4.6 Define attendance report generation

5. Data Management & Security

- 5.1 Define database tables and relationships
- 5.2 Define facial data storage method
- 5.3 Define access control rules
- 5.4 Define audit logging requirements
- 5.5 Define data retention policies

6. Testing & Validation Planning

- 6.1 Create requirement-based test cases
- 6.2 Define system acceptance criteria
- 6.3 Plan usability testing
- 6.4 Plan performance testing
- 6.5 Define accuracy evaluation metrics

7. Documentation & Delivery

- 7.1 Create user manual
- 7.2 Create installation/setup guide
- 7.3 Prepare system documentation
- 7.4 Prepare final presentation materials
- 7.5 Final demonstration preparation